

SANUTH MANDEPA

Software Engineering Undergraduate

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PROFILE

As a final-year Software Engineering student, I am passionate about turning complex problems into seamless, reliable user experiences. Through a year of industry experience in web design and hands-on work building independent full-stack projects, I have developed a versatile technical foundation encompassing UI/UX design, backend logic, machine learning integration, and containerized deployment. I thrive in collaborative agile environments, love picking up new tools, and take pride in seeing a project through from initial concept to a successful, stable launch. Ultimately, my greatest assets are my adaptability and a strong, continuous drive to learn. I am eager to bring this proactive, problem-solving mindset to a dynamic engineering team where I can contribute across the development lifecycle and rapidly grow.

SKILLS

- **Programming & Scripting:** Python, JavaScript, Java, PHP, SQL, Bash (Basic), HTML, CSS
- **System Testing & Quality Assurance:** Manual QA Testing, Issue Documentation, UI Validation
- **DevOps & Tools:** Docker, Google Cloud Run, Git, GitHub, Linux (fundamentals), CI/CD (concepts), AWS (Basic), Postman
- **Frameworks & Libraries:** React.js (Basic), Flask, Node.js, Tailwind CSS, Bootstrap, PyTorch
- **AI & Applied Machine Learning:** OpenAI API, LangChain, AI-Assisted Workflows, ML Pipeline Design, Deep Learning
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Design & UI/UX:** Figma, Adobe Photoshop, Adobe After Effects, GSAP, WordPress
- **Soft Skills:** Team Leadership, Strong Problem-Solving & Analytical Thinking, Proactive & Collaborative Mindset, Attention to Detail, Ability to Take Ownership, Continuous Learning

PROFESSIONAL EXPERIENCE

Intern Web Designer

August 2024 to August 2025

Weblook International PVT Limited Colombo 07, Sri Lanka

- Contributed to the design, implementation, and iteration of responsive production websites, collaborating closely with developers to deliver user-facing features.
- Created wireframes, prototypes, and UI flows using Figma, translating designs into live WordPress implementations using Elementor, Divi, and WooCommerce.
- Performed quality assurance (QA) testing on live websites, documented issues, and implemented fixes to improve usability and responsiveness.
- Supported post-deployment updates by making controlled changes to live sites while maintaining design consistency and accessibility standards.
- Assisted with interactive animations and visual enhancements using GSAP, along with video editing and design assets created using After Effects and Photoshop.
- Contributed to the design, implementation, and iteration of responsive production websites
- Shared knowledge with fellow interns and collaborated effectively within a cross-functional team environment.

PROJECTS

ChagaSight: A Self-Supervised Dual-Pathway Vision Transformer Ensemble for Chagas Disease Detection from 12-Lead ECG ([GitHub Repo](#))([Live Demo](#))

August 2025 - Ongoing

Stack: Python, PyTorch, Flask, React.js, Tailwind CSS, Docker, WFDB, HuggingFace Spaces

- Designed a 173M-parameter hybrid deep learning system combining 1D temporal and 2D spatial Vision Transformer pathways with REPA cross-modal alignment for Chagas disease screening from 12-lead ECG recordings.
- Engineered a modular automated preprocessing pipeline covering bandpass filtering, resampling, per-lead normalisation, and 2D ECG image construction using NumPy and SciPy.
- Implemented self-supervised pretraining (MAE + ST-MEM) on ~366,000 unlabeled ECG recordings across three datasets (PTB-XL, CODE-15%, SaMi-Trop), followed by 5-fold stratified cross-validation with ensemble averaging.
- Achieved AUROC of 0.8707 and TPR@5% of 0.4958 on the 5-fold hybrid ensemble, outperforming single-pathway baselines (1D: AUROC 0.8567, 2D: AUROC 0.7079).
- Containerised and deployed a Flask REST API using Docker to HuggingFace Spaces and a React.js frontend to Vercel, supporting real-time ECG upload and inference across three model modes (1D, 2D, Hybrid).

Internova: Coding and Mock Interview Preparation Platform ([GitHub Repo](#))

September 2023-June 2024

Stack: Python, Flask, REST APIs, LangChain, OpenAI API, Docker, Kubernetes, Google Cloud Run, GitHub, React.js, Node.js, MongoDB

- Led a student development team in building a full stack web platform supporting coding practice and mock interview preparation.
- Gained introductory exposure to CI/CD concepts through build and deployment experimentation on Google Cloud Run.

- Applied Docker containerisation to package and deploy the Flask backend service, improving environment consistency across development and deployment stages.
- Designed and implemented a speech emotion analysis service using Python and Flask to analyse user audio responses.
- Developed a machine learning inference pipeline using MFCC feature extraction and a trained classification model.
- Built REST API endpoints to process uploaded audio and return predicted emotional states.
- Integrated a Retrieval Augmented Generation component to provide contextual feedback to users.
- Collaborated using Git and GitHub within an agile team environment.

Online Shopping System GUI Application

October 2023-November 2023

Stack: Java, Java Swing

- Developed a desktop based shopping management system using object oriented programming principles
- Implemented product tracking and user interface functionality

Theatre Seat Management System Command Line Application

February 2023-March 2023

Stack: Java

- Built a command line application to manage theatre seat allocation and session control

University Progression Predictor

October 2022-November 2022

Stack: Python

- Developed a Python based command line tool to predict student progression outcomes based on academic credit inputs

EDUCATION

University of Westminster (UK) Delivered by Informatics Institute of Technology Sri Lanka

2022 - Present

BEng(Hons) Software Engineering

Ananda College, Sri Lanka

2012 - 2021

G.C.E. Advanced Levels Biological Science Stream

CERTIFICATIONS

- AWS Educate Introduction to Cloud 101 Issued by Amazon Web Services [Certificate](#)
- Version Control Issued by Meta [Certificate](#)
- Postman API Fundamentals Student Expert [Certificate](#)
- Introduction to Generative AI Issued by Google [Certificate](#)
- What Is Generative AI Issued by LinkedIn Learning [Certificate](#)
- Using Python to Access Web Data Issued by University of Michigan via Coursera [Certificate](#)
- Python Data Structures Issued by University of Michigan via Coursera [Certificate](#)
- Programming for Everybody Getting Started with Python Issued by University of Michigan via Coursera [Certificate](#)
- Foundations of Project Management Issued by Google [Certificate](#)
- The Fundamentals of Digital Marketing Issued by Google Digital Garage [Certificate](#)
- React.js Essential Training Issued by LinkedIn Learning [Certificate](#)
- PHP Essential Training Issued by LinkedIn Learning [Certificate](#)
- Start the UX Design Process Empathize Define and Ideate Issued by Google via Coursera [Certificate](#)
- Introduction to DevOps Issued by IBM via Coursera Ongoing
- Financial Markets Issued by Yale University via Coursera Ongoing

INTERESTS

- Continuous learning
- Badminton
- Self improvement
- Video editing
- Music

REFERENCES

Mr. Indrajith Ekanayake

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Mr. Manul Singhe

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